

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 3 – CLASS TEST (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 3 – CLASS
Weightage:	35% of CIA	Total Marks:	TEST (5 Questions × 10 Marks)
CO1 Statement:	Apply the fundamental concepts of machine learning and neural networks to analyze learning algorithms using perceptrons, multilayer perceptrons, forward backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.		
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Section / Batch:	2024	Date:	11-08-2026

Code of Conduct

I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: CH. Durga Prasad

QUESTIONS: 5 Total Marks: 50

Class Test

Duration: 1hr

1. A company developing an AI-based disease diagnosis system is deciding whether to use a traditional Machine Learning model or a Deep Learning model. Compare both approaches by discussing their feature extraction methods, data requirements, computational needs, and suitability for image-based medical diagnosis.

1. Traditional Machine Learning vs Deep Learning for Disease Diagnosis

Aspect	Traditional Machine Learning	Deep Learning
Feature Extraction	Features are manually designed by experts, such as edges, texture, shape, and intensity.	Features are automatically learned from raw images through multiple neural network layers.
Data Requirement	Requires comparatively less training data.	Usually requires a large amount of labeled data.
Computational Needs	Lower computational requirements; can run on normal CPUs.	High computational requirements; GPUs/TPUs are commonly used.
Accuracy	Good for structured and simpler problems.	Generally performs better for complex image-based tasks.
Medical Images	Requires expert knowledge for feature engineering.	Highly suitable for X-rays, CT scans, MRI, skin images, etc.

Conclusion:

For an AI-based disease diagnosis system using medical images, **Deep Learning is generally more suitable** because it automatically learns complex features and can achieve high accuracy when sufficient training data and computational resources are available.

2.A facial recognition application receives raw image data without any manually extracted features. Explain how Representation Learning enables the model to automatically learn useful features from the images and discuss its advantages over manual feature engineering.

Representation Learning in Facial Recognition

Representation Learning is a technique where a neural network automatically learns useful representations or features directly from raw input data.

For facial recognition:

Raw Image → Neural Network → Feature Learning → Face Representation → Classification/Matching

For example, the network may automatically learn:

- Early layers → edges and basic shapes
- Middle layers → eyes, nose, mouth, and facial patterns
- Deeper layers → complete facial characteristics and identity-related features

Advantages over manual feature engineering:

1. **Automatic feature extraction:** No need to manually identify facial features.
2. **Better accuracy:** Deep networks can learn complex patterns that humans may miss.
3. **Works with raw data:** Images can be directly provided as input.
4. **Scalable:** The same approach can learn features for large and diverse datasets.
5. **Adaptable:** Features can improve automatically as the model is trained with more data.

Conclusion:

Representation Learning reduces the need for human-designed features and allows the model to learn meaningful facial representations automatically from raw images.

3. Two neural network models are designed for an image classification task. One model contains many hidden layers with fewer neurons, while the other has a single hidden layer with a large number of neurons. Compare these architectures and discuss how width and depth influence learning performance.

The two architectures differ mainly in depth and width.

- Depth = Number of hidden layers.
- Width = Number of neurons in each layer.

Feature	Deep & Narrow Network	Shallow & Wide Network
Hidden layers	Many	Few
Neurons per layer	Fewer	More
Feature learning	Learns hierarchical features	Learns fewer levels of abstraction
Complexity	Can represent complex patterns	Can represent complex patterns but less hierarchically
Training	May be harder	Usually simpler
Image tasks	Generally more suitable	May be less effective for complex images

A deep network can learn hierarchical representations such as:

Edges → Shapes → Parts → Objects

A wide network has many neurons in a layer and can learn many patterns at the same level.

Conclusion:

For complex image classification, depth is usually more beneficial because it enables hierarchical feature

learning. However, excessive depth can cause problems such as vanishing gradients and higher computational cost. A suitable balance between width and depth gives better performance.

4. A Deep Neural Network experiences slow learning due to the vanishing gradient problem during training. Compare the Sigmoid and Tanh activation functions and explain their impact on gradient flow, convergence speed, and overall network performance.

Sigmoid vs Tanh and the Vanishing Gradient Problem

Feature	Sigmoid	Tanh
Output range	0 to 1	-1 to +1
Zero-centered	No	Yes
Gradient	Can become very small	Generally stronger around zero
Vanishing gradient	More severe	Less severe than Sigmoid
Convergence	Slower in deep networks	Usually faster than Sigmoid
Common use	Binary output layer	Hidden layers in some networks

Sigmoid function:

Tanh function:

When Sigmoid or Tanh receives very large positive or negative inputs, their gradients become very small. During backpropagation, these small gradients are multiplied through many layers, causing the **vanishing gradient problem**. Tanh is generally better than Sigmoid for hidden layers because it is **zero-centered** and has a stronger gradient around zero.

Conclusion:

Between Sigmoid and Tanh, **Tanh is generally preferable for hidden layers** because it provides better gradient flow and faster convergence. However, modern deep networks commonly use **ReLU-based activations** to further reduce vanishing-gradient issues.

5.An image classification model suffers from inactive neurons during training, leading to poor prediction accuracy. Evaluate the use of ReLU, Leaky ReLU, and ELU activation functions for this problem and recommend the most suitable choice with justification.

ReLU vs Leaky ReLU vs ELU for Inactive Neurons

Inactive neurons are commonly caused by the dying ReLU problem, where ReLU outputs zero for negative inputs and a neuron may stop learning.

Activation	Formula	Main Advantage	Problem
ReLU	$\max(0,x)$	Simple and fast	Can produce dead neurons
Leaky ReLU	$x \text{ if } x>0,$ αx otherwise	Allows small negative gradients	Requires choosing α
ELU	$x \text{ if } x>0,$ $\alpha(e^x-1)$ otherwise	Smooth negative region and better gradient flow	More computationally expensive

ReLU:

For negative values, the output and gradient are zero. Therefore, neurons can become permanently inactive.

Leaky ReLU:

It allows a small gradient for negative inputs, helping prevent inactive neurons.

ELU:

ELU provides a smooth negative region and can improve learning and convergence.

Recommendation:

For the given problem, Leaky ReLU is the most suitable direct replacement for ReLU because it allows negative inputs to produce a small non-zero gradient, reducing the dying-neuron problem while remaining computationally simple.

Overall conclusion:

ReLU → simple but may create dead neurons

Leaky ReLU → prevents dead neurons effectively

ELU → better smoothness but slightly more computationally expensive.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge Course Coordinator Program Director

Signature: _____ Signature: _____ Signature: _____ Date: _____

_____ Date: _____ Date: _____